GRADE 7 MATHEMATICS SPOT TEST

Duration: 20 Minutes **Total Marks:** 20 Marks

Name: ..... **Class/Index:** .....

Part A: Directed Numbers (Questions 1–5)

1. Fill in the blank: `(+8) + (-3) = (.....)` [1 mark]
2. Fill in the blank: `(-6) + (+11) = (.....)` [1 mark]
3. The ground floor of a building is named Floor 0, floors above are positive (1, 2, 3...), and floors below are negative (-1, -2, -3...). If a person in **Floor 5** climbs up a further **4 floors**, which floor will he be in? [1 mark]
4. If a person in **Floor 3** descends a further **5 floors**, which floor will he be in? [1 mark]
5. The temperature at 6:00 a.m. in a city was -3.5°C, and by afternoon it increased by 8°C. Find the temperature in the afternoon. [1 mark]

Part B: Time (Questions 6–10)

6. Complete the addition:

	Years	Months	Days
	5	4	12
+	2	3	10
	(____)	(____)	(____)

[1 mark]

7. Complete the subtraction:

	Years	Months	Days
	6	8	20
-	2	4	10
	(____)	(____)	(____)

[1 mark]

8. Write down the decade to which the year **AD 1974** belongs. [1 mark]
9. Write down the century to which the year **AD 1815** belongs. [1 mark]
10. A teacher started work in a school on **2001 - 07 - 13** and was transferred on **2015 - 11 - 22**. Find the total period of service in years, months, and days. (Format: Years - Months - Days) [1 mark]

Part C: Indices (Questions 11–15)

11. Write down using index notation: $m \times m \times m \times n \times n$ [1 mark]
12. Write down using index notation: $3 \times 3 \times 3 \times p \times p$ [1 mark]
13. Expand and write as a product: x^3 [1 mark]
14. Expand and write as a product: $2p^2$ [1 mark]
15. Expand and write as a product: 3^2y^3 [1 mark]

Part D: Algebraic Expressions (Questions 16–20)

16. Write the expression in words: $10 + (x / 2)$ [1 mark]
17. The length of a room is y meters less than twice its width. The width of the room is $4\ m$. Write an expression in terms of y for the length of the room. [1 mark]
18. Simplify the following expression: $5a + 4b - 2a - b$ [1 mark]
19. Find the value of the expression: $4x - 3y$ when $x = 5$ and $y = 2$. [1 mark]
20. A father's age was 30 years at the time his son was born. Write the age of the father in terms of x when his son is x years old. [1 mark]